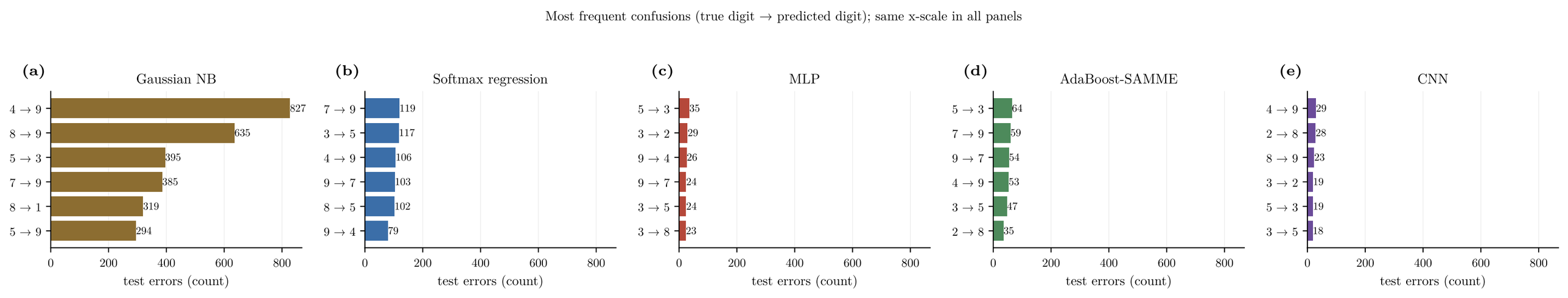
Most frequent confusions (true digit → predicted digit); same x-scale in all panels
(a)
Gaussian NB
4 → 9
8 → 9
5 → 3
7 → 9
8 → 1
5 → 9
827
635
395
385
319
294
0
200
400
600
800
test errors (count)
(b)
Softmax regression
7 → 9
3 → 5
4 → 9
9 → 7
8 → 5
9 → 4
119
117
106
103
102
79
0
200
400
600
800
test errors (count)
(c)
MLP
5 → 3
3 → 2
9 → 4
9 → 7
3 → 5
3 → 8
35
29
26
24
24
23
0
200
400
600
800
test errors (count)
(d)
AdaBoost-SAMME
5 → 3
7 → 9
9 → 7
4 → 9
3 → 5
2 → 8
64
59
54
53
47
35
0
200
400
600
800
test errors (count)
(e)
CNN
4 → 9
2 → 8
8 → 9
3 → 2
5 → 3
3 → 5
29
28
23
19
19
18
0
200
400
600
800
test errors (count)